

Meteor Trail Characteristics Relevant to Meteor Burst Communications Observed by Two Meteor Radars in Yunnan, Southwest China

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Abstract: Meteor-burst communications (MBC) use ionized meteor trails to enable long-distance, beyond-line-of-sight links. Most existing models address long-range forward-scatter configurations, whereas back-scatter scenarios for short-range communications are less well studied. This paper presents a statistical, comparative analysis of meteor-trail parameters using observations from two back-scatter meteor radars operating at different frequencies (53.1 MHz at Qujing, 25.6°N, 103.8°E; and 39.0 MHz at Lincang, 23.63°N, 99.83°E) in Yunnan, Southwest China. The analysis is based on data collected over a 46-day period from 13th December 2024 to 27th January 2025, during which the 39.0 MHz system detected approximately 360 000 meteor events and the 53.1 MHz system detected approximately 280 000 events. We focus on spatiotemporal characteristics, decay times, and inter-arrival times of meteor trails. Detection rates exhibit strong diurnal variation, with higher counts in the local morning. Operating frequency strongly affects the detection of specular height: the peak of

the meteor specular height distribution occurs near 85 km for the 53.1 MHz system and near 88 km for the 39.0 MHz system. While the lower-frequency radar generally provides higher echo occurrence rate and shorter inter-arrival times, this advantage reverses during local afternoon hours, when the higher-frequency system achieves a higher echo occurrence rate coupled with shorter inter-arrival times. These results provide quantitative constraints and prior information useful for optimizing short-range MBC systems, particularly with respect to frequency selection, temporal scheduling, and antenna configuration.

Keywords: channel characterization; meteor burst communications; meteor trail statistics; radar observation; time-space distribution; underdense trail

I. INTRODUCTION

Meteor burst communications (MBC) are a beyond-line-of-sight communications technology that leverages ionized meteor trail for long-distance signal trans-

mission [1]. When the meteors burn at the altitudes of 70–120 km, the ionized meteor trail is leaving behind [2]. MBC utilizes the meteor trail to reflect radio signals, enabling reliable medium to long-range transmissions from 500 to 2000 km. MBC systems have several distinct advantages. Their compact hardware footprint contributes to lower operational costs [3]. Furthermore, the technology also delivers strong anti-jamming performance and high survivability [4]. Therefore, MBC has more rapid blackout recovery [5]. It also provides a relatively new and uncongested means to conventional channels [6]. It allows their use in polar regions and in unpopulated areas [7]. So MBC stands out as a promising solution to offering critical advantages for emergency communications in remote regions and for military operations.

A critical aspect of MBC system design is accurate channel modeling, which relies on precise statistics of meteor trail characteristics. The vast majority of the current MBC systems simulation models are based on the forward-scatter mode [5, 8–12]. The forward-scatter models rely on transmitters and receivers that are separated by a considerable distance. This is similar to the forward-scatter meteor radars [13]. These models employ a simplified mathematical approach that incorporates key assumptions [7]. For example, they assume the distribution and variance of meteor trails [14], and assumptive meteor trails lie midway between transmitter and receiver [15]. However, the results of the forward-scatter meteor radar indicate that meteor trails do not only occur in this direction [16].

These forward-scatter models have another problem. Weitzen et al. pointed out that many existing models are not effective for meteor burst communications at short ranges, which in this context refers to distances below approximately 400 km [17]. In this case, the propagation may use both forward-scatter and back-scatter. Sometimes they use only back-scatter [8]. Lapshina et al. also reported the occurrence of a type of meteor trail termed forward backward scattering (FBS), referring to such meteor trails simply as FBS trails, this type of meteor trails might provide precise time synchronization [18]. Back-scatter meteor radars share similarities with the FBS mechanism. A back-scatter radar works from a single site. It detects signals reflected directly back from the meteor trail. This provides continuous local data. So, data from back-scatter radars can improve MBC models [19].

While meteor radar observations are a direct source of the statistical data needed for MBC channel characterization, the focus of most meteor radars has been on atmospheric science, particularly wind field retrieval in the mesosphere and lower thermosphere (MLT) region [20]. Meteor radars predominantly operate in the very high frequency (VHF) band, with a typical frequency range of approximately 30 MHz to 50 MHz. This frequency range aligns with the simulated transmission frequencies used in numerous studies [1, 5, 7, 21]. These radars are also capable of measuring key trail parameters such as slant range, azimuth, zenith angle, decay time, and specular height [22].

Furthermore, meteor radars can detect different types of meteor trail echoes, including both underdense and overdense trail [23]. These two types of meteor trail are classified based on their electron line density, with a critical value of approximately 10^{14} electrons per meter. Meteor trail with an electron line density below this threshold are classified as underdense, whereas those exceeding it are termed overdense [13]. MBC predominantly utilize underdense trail [24]. Several key physical parameters of meteor trail for MBC system design can be directly provided by meteor radar observations [25]. Consequently, meteor radars can provide statistical data to support the optimization of MBC systems. Despite this potential, a systematic, observationally grounded analysis of meteor trail parameters specifically suited for short-range MBC channel modeling remains lacking. Existing statistical studies are predominantly derived from forward-scatter models or radar data not optimized for this specific application. Consequently, a significant gap persists in exploiting concurrent, long-term observations from back-scatter meteor radars, whose configuration is most analogous to short-range MBC links, to rigorously quantify key channel parameters and their dependence on critical factors such as operating frequency.

This study aims to address this gap by presenting a comprehensive statistical analysis of meteor trail characteristics using concurrent observations from two back-scatter meteor radars operating at different frequencies (39.0 MHz and 53.1 MHz) in Yunnan, Southwest China. We focus on the parameters most critical for short-range MBC channel modeling: The spatiotemporal distribution of trail occurrence, specular height, slant range, decay time (linked to usable burst duration), inter-arrival time (linked to channel availability),

and signal-to-noise (SNR). By comparing results across two frequencies, we directly assess a key system design variable. Our analysis provides quantitative, empirical constraints on these parameters, offering essential prior information and benchmarks for developing more accurate channel models and optimizing system design — particularly in terms of frequency selection, temporal scheduling, and antenna configuration — for short-range meteor burst communications.

The remainder of this paper is organized as follows. Section II describes the two meteor radar systems, including their hardware configurations and the signal processing methodologies used to retrieve key meteor trail parameters relevant to MBC channel modeling. Section III presents the observational results regarding the spatiotemporal distribution, slant range, and specular height of meteor trails. Section IV provides a detailed analysis of decay time and inter-arrival time characteristics, discussing their implications for channel availability and MBC system design. Section V examines the SNR characteristics of meteor echoes, analyzing their variability and impact on link quality and system performance. Finally, Section VI summarizes the principal findings of this study and outlines prospective directions for future research.

II. METEOR RADARS CONFIGURATIONS AND SIGNAL PROCESSING FOR CHANNEL PARAMETER RETRIEVAL

The data employed in this paper were obtained from two meteor radar observations, including the Qujing Meteor Radar (QJMR) [26] and the Shuangjiang Meteor Radar (SJMR). QJMR is located at the Electromagnetic Environment Observation Station in Qujing, Yunnan Province (25.6°N, 103.8°E). SJMR is situated at Shuangjiang (23.63°N, 99.83°E), both in Yunnan Province. The distance between these two stations is 480 km.

QJMR is a stratosphere-troposphere/meteor radar (ST/MR) operating at 53.1 MHz, with the system configuration as per Reid et al. [27]. The SJMR is a specular meteor radar system similar with the KMMR. The SJMR is a specular meteor radar system operating at 39.0 MHz, which is similar with the QJMR. They all operating in back-scatter mode, and utilize a configuration in which the transmitter and receiver are co-located. This arrangement enable autonomous operation and in-

dependent detection.

2.1 Estimation of Angle of Arrival

Finding the direction of meteor trails reveals estimating the azimuth and zenith. In meteor radar, the antenna array is used for direction finding of meteor trails. The Jones antenna configuration [28] is a popular antenna array in most meteor systems. The Jones configuration is a five-antenna interferometric antenna array, where the antennas aligned along the same line with the central antenna form an interferometric baseline. The two meteor radar systems used in this paper differ in the angle between their baselines and true north. Figure 1 illustrates the antenna array layout of the SJMR and QJMR system. The arrival angle of an incoming wave is accurately determined by measuring the phase difference of the same incident wave received by two antennas positioned along such a baseline.

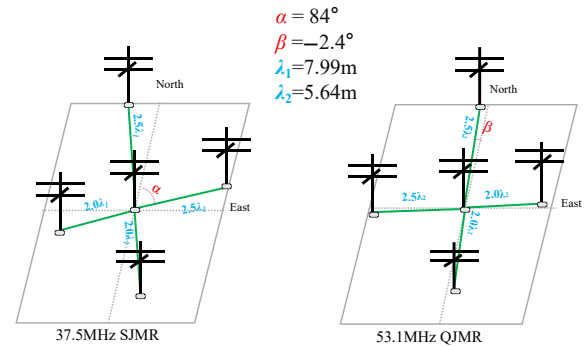


Figure 1. The Jones configuration interferometric antenna array implemented in the SJMR (left) and QJMR (right) system, comprising one transmitting antenna (TX) and five receiving antennas (RX1–RX5) [29].

When a wavefront with wavelength λ and arrival angle α is received by two antennas, the phase difference χ between the signals at the two antennas is given by

$$\chi = 2\pi \frac{d}{\lambda} \sin \alpha, \quad (1)$$

where d is the distance between the two antennas. By measuring the phase difference χ , the arrival angle α of the incident wave can be determined. However, a unique solution for α is only achievable when the antenna baseline distance $d \leq 0.5\lambda$. In practice, when antennas are placed too close, mutual coupling between them can introduce significant errors in the phase difference estimation. While increasing the antenna spacing

can mitigate mutual coupling, it introduces ambiguity in the arrival angle solution.

To resolve this issue, the Jones configuration dexterously arranges three antennas along the same baseline with spacings d_1 and d_2 such that $|d_1 - d_2| \leq 0.5\lambda$. This configuration ensures a unique solution for the arrival angle α . In the radar systems discussed in this paper, the three antennas on the same baseline are spaced at 2.0λ , 2.5λ and 4.5λ with measured phase differences $\chi_{2.0\lambda}$, $\chi_{2.5\lambda}$ and $\chi_{4.5\lambda}$ respectively.

$$\chi_{2.0\lambda} = -\frac{2\pi d_1}{\lambda} \sin \alpha, \quad (2)$$

$$\chi_{2.5\lambda} = +\frac{2\pi d_2}{\lambda} \sin \alpha, \quad (3)$$

$$\sin \alpha = -\frac{\lambda}{2\pi} \frac{(\chi_{2.0\lambda} - \chi_{2.5\lambda})}{(d_1 + d_2)}, \quad (4)$$

$$\sin \alpha = -\frac{\lambda}{2\pi} \frac{(\chi_{2.0\lambda} + \chi_{2.5\lambda})}{(d_1 - d_2)}. \quad (5)$$

Based on the above formulations, the accurate and unambiguous arrival angles α_1 and α_2 for each antenna baseline can be obtained. Using these angles, the azimuth angle φ and the zenith angle θ of the meteor can be determined by

$$\cos^2 \theta = 1 - \cos^2 \alpha_1 - \cos^2 \alpha_2, \quad (6)$$

$$\varphi = \cos^{-1}(\alpha), \quad (7)$$

where α is the angle of arrival measured by the baseline corresponding to an azimuth of 0° . It is noteworthy, the true north direction corresponds to an azimuth angle of 0° in meteor radar systems.

2.2 Determination of Meteor Trail Specular Height and Slant Range

This subsection describes the methodology employed by the meteor radar system for calculating the slant range and specular height of meteor trails. The slant range refers to the distance from the meteor trails to the receiving antenna. It is derived from the time delay of the received signal.

$$R = c \cdot \tau, \quad (8)$$

where c is the speed of light, and τ is the receiver's time delay, and R denotes the measured slant range detected by the radar. However, an inherent slant range ambiguity exists is attributable to the echo delay is measured relative to the radar's transmitted pulse train. The procedure for resolving this ambiguity is described as follows.

$$R_i = R + iR_{\text{amb}}, \quad i = 0, \dots, N, \quad (9)$$

where R_{amb} denotes the slant range ambiguity limit, it is given by $R_{\text{amb}} = c/(2 \cdot \text{PRF})$. The PRF is the pulse repetition frequency. The two meteor radar systems considered in this paper are operated with PRF = 430 Hz, and the slant range ambiguity is approximately 348.9 km. The correct value of i is determined by ensuring the computed meteor trails specular height falls within the typical altitude range of 70 km to 110 km [22], thus i is usually 0. The formula for calculating meteor trails specular height is

$$h = \sqrt{R_e^2 + R_i^2 + 2R_i R_e \cos \theta} - R_e, \quad (10)$$

where R_e is the radius of the Earth, h is the meteor trails specular height above the Earth's surface, and i usually is 0.

2.3 Decay Time of Meteor Trails

In the meteor radar systems, the decay time is used to calculate the ambipolar diffusion coefficient [30]. It is defined as the time taken for the echo power to decay below a predefined threshold. This parameter characterizes the persistence of a meteor trails echo. Although MBC systems lack a precise definition of meteor trails duration, the two concepts still exhibit similarities [1]. Without loss of generality, an example using QJMR is presented below to illustrate the decay time calculation process in meteor radar systems.

Figure 2 shows the typical echo power and phase profiles of an underdense meteor trail observed from a randomly selected data record by QJMR on 14th December 2024. The power profile exhibits a rapid rise to peak amplitude followed by exponential decay. The phase profile demonstrates a linear progression from the peak point to the end point, indicating stable phase

variation during the meteor trail's decay process. Three points are defined as the start time (t_1), peak time (t_2) and end time (t_3). Decay time equals the time between t_2 and t_3 . Decay time is calculated by the formula as follows,

$$T_{dt} = \delta(t_3 - t_2), \quad (11)$$

where δ is the sampling interval, and in the case of QJMR and SJMR used in this paper, $\delta = 0.0093023255$ s in the MRs. However, meteor radar systems apply multiple criteria to determine whether detected meteor trails are suitable for atmospheric wind field retrieval and other geophysical inversions. A detailed classification of these screening criteria can be found in Holdsworth et al. [22]. For example, these criteria include constraints on the echo rise time—defined as the time interval between the trails start time (t_1) and the peak time (t_2).

III. SPATIOTEMPORAL STATIC DISTRIBUTION

This section presents observational results from the MRs. A comparative analysis of the QJMR and SJMR investigates the frequency-dependent characteristics of meteor trail parameters in time and space. To ensure a consistent and meaningful comparison, both stations utilize data from the same time period, spanning from 13th December 2024 to 27th January 2025. Unless otherwise specified, all results presented hereafter correspond to this observation period.

Figure 3(a) illustrates an example of the meteor counts separately detected by the two MRs, both the MRs detected significant daily variability in meteor trails detection counts. Despite these fluctuations, the daily meteor counts remained relatively stable throughout the observation period. On average, the SJMR recorded 7842 meteor trails per day, while the QJMR detected an average of 6153 meteor trails per day. Figure 3(b) shows that the hourly meteor counts exhibit a significant non-uniform distribution. The maximum meteor count was reached in the local morning, while the fewest meteors were detected in the local afternoon. This observation is consistent with the findings reported by Zeng et al., and they also present the causes, the difference in echo occurrence rate is primarily attributed to the different operating frequencies of the

two meteor radars [16]. For SJMR and QJMR, the operation frequency of SJMR is 39.0 MHz, while it is 53.1 MHz for QJMR. The higher operating frequency of the SJMR enables it to detect a greater meteor detection event counts compared to the QJMR system. These statistics provide crucial observational constraints for refining physics-based channel models in MBC. They are instrumental in accurately simulating parameters such as temporal channel availability and trail occurrence probability, which directly impact the design and performance prediction of MBC systems.

Figure 3(c) presents the slant range distribution of meteor trails. The meteor radars detect most meteor trails within a slant range of 100 km to 300 km from the receiving antenna. The median slant range of SJMR and QJMR is 144 km and 153 km respectively. In the back-scatter radar systems, the total path of electromagnetic wave propagation is twice the distance. For back-scatter meteor radar, estimating the time from transmitted pulse to acquisition of meteor trails physical parameters require twice the slant range. Based on the most meteor slant range distribution, the time delay for meteor radar receivers to acquire physical parameters falls within approximately 0.66–2 ms. This shares similarities with the approach proposed by Lapshina et al. [18], which utilizes FBS to achieve clock synchronization.

Figure 3(d) shows the specular height distribution of meteor trails. The peak specular heights of the MRs are 87.8 km for SJMR and 85.3 km for QJMR separately. The specular height of the maximum meteor echo counts for the QJMR is lower than the SJMR, this is attributable to the meteor specular height distributions detected by lower-frequency meteor radars are characteristically higher than those detected by higher-frequency meteor radars. These results suggest that MBC channel modeling should take into account the actual operating frequency of the transmitter. Knowing the slant range and specular height distribution of meteor trails enables more accurate estimation of signal link loss.

Figure 3(e) and (f) show the angle of arrival (zenith and azimuth) of meteor echo observed by the SJMR and QJMR. The angular distributions observed by both MRs are generally similar. For the zenith angles, SJMR are distributed between 20° and 78° , which is similar with QJMR. For the azimuth angles, the two MRs show a relatively uniform distribution. However, SJMR

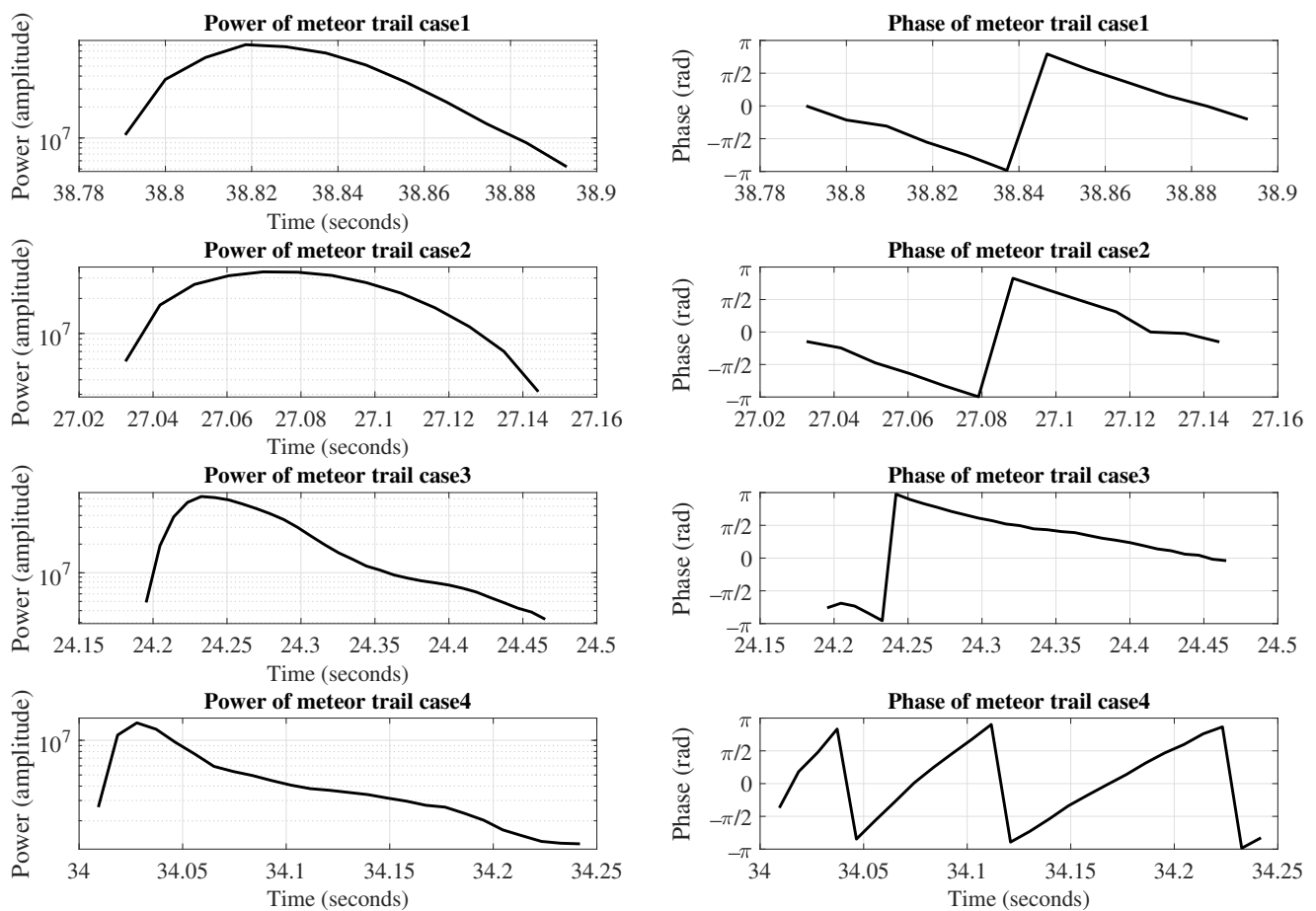


Figure 2. The echo power (left) and phase (right) profiles of meteor trail cases observed from four randomly selected data records by QJMR on 13th December 2024. Each meteor event is shown from start to end.

shows a reduced echo occurrence rate between 30° and 75° . These regions of lower meteor density are consistent with the distributions shown in Figure 4(a).

Figure 4 shows the scatter and density distribution of meteors. SJMR exhibits reduced detections in the north-east direction. This is attributed to a wall located near the transmitting antenna at the SJMR site. It obstructs the signal in these azimuthal sectors.

IV. DECAY TIME AND INTERVAL TIME CHARACTERISTICS OF METEOR TRAILS

Building upon the spatiotemporal distribution established in the preceding chapter, this section focuses on two critical parameters in MBC systems. They are respectively decay time and inter-arrival time. These factors directly control the availability and persistence of communication links. The overall distributions are presented in Figure 5. Both parameters exhibit highly

similar distributions. For the decay time shown in the Figure 5(a), the distributions of SJMR and QJMR are remarkably similar, with identical median values of 0.10 s, indicating that most meteor trails persist for only a very short duration. Similarly, the inter-arrival time distributions, illustrated in Figure 5(b), also exhibit highly concentrated profiles, with the majority of intervals being shorter than 50 s. The median inter-arrival time is slightly longer for the QJMR (7.04 s) than the SJMR (6.93 s).

Table 1 presents the cumulative distribution of meteor trail decay time and inter-arrival time for the SJMR and QJMR radar systems from 13th December 2024 to 27th January 2025. For decay time, the median value at the 50th percentile is identical for both radars at 0.10 s. QJMR generally exhibits slightly shorter decay times than SJMR. This systematic difference can be attributed to the higher operating frequency of QJMR. The results further show that over 99% of all detected meteor trails persist for less than 0.70 s. For inter-arrival

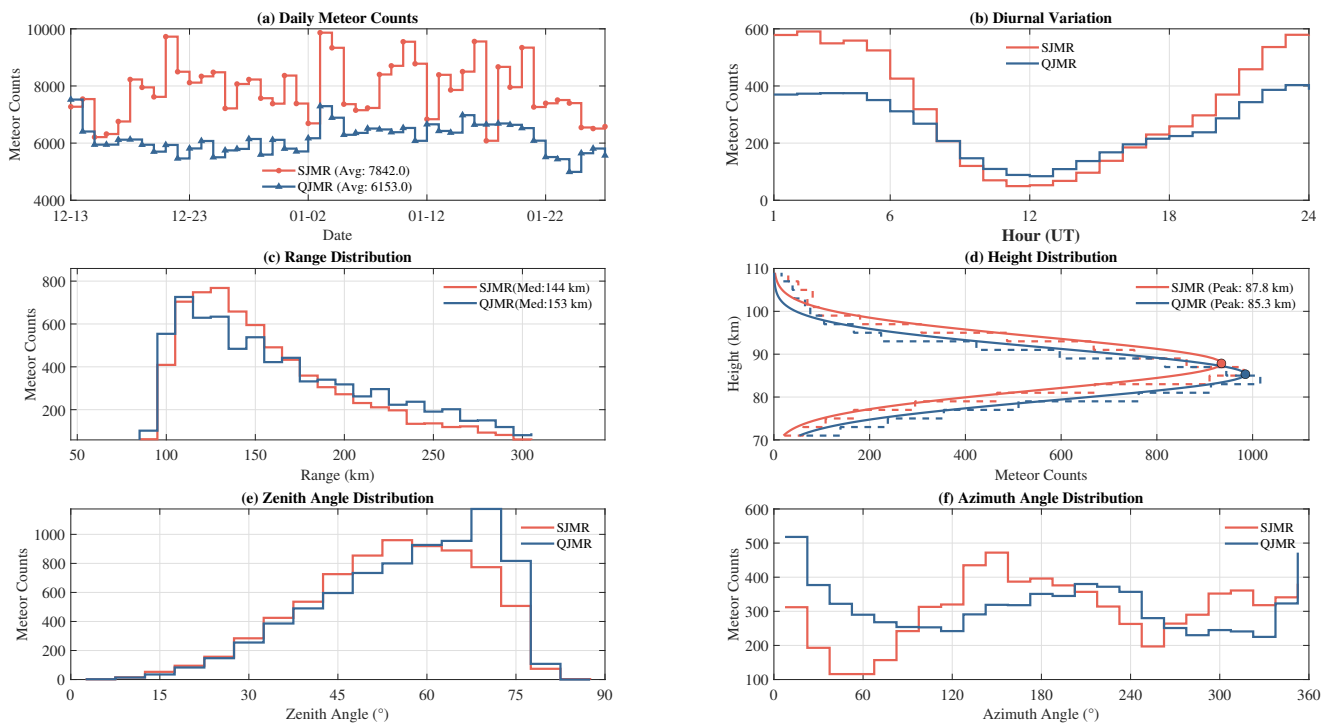


Figure 3. Statistical distributions of key meteor trail parameters observed by the QJMR and SJMR from 13th December 2024 to 27th January 2025. (a) Daily meteor counts showing the temporal variability of meteor echo occurrence rate (SJMR average: 7842.0; QJMR average: 6153.0). (b) Diurnal variation of meteor occurrence, peaking during local morning hours. (c) Slant range distribution, showing that meteor trails detected by QJMR are generally at longer distances (median: 153.0 km) compared to SJMR (median: 144.0 km). (d) Specular height distribution, revealing a distinct difference in the peak altitude of meteor trails, with SJMR peaking at 87.8 km and QJMR at 85.3 km, and the variance is 44.0 and 43.4 separately. (e) and (f) Zenith and Azimuth angle distribution, showing the directional characteristics of meteor trail detections.

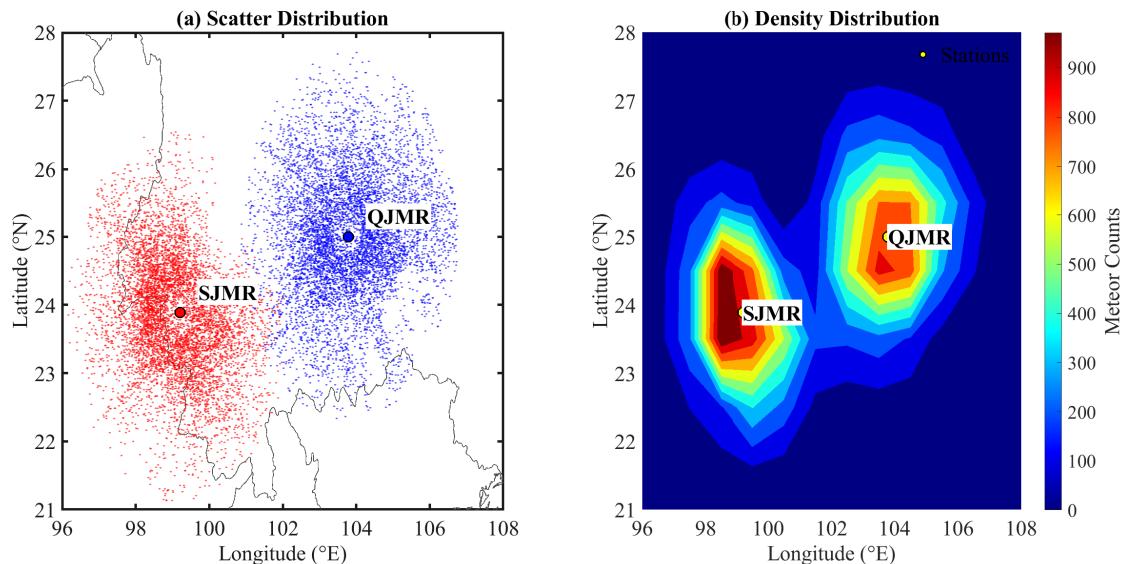


Figure 4. Distributions of meteors detected by the QJMR and SJMR with respect to horizontal location from 13 December 2024 to 27 January 2025. (a) Scatter distribution. (b) Density distribution.

time, SJMR exhibits shorter intervals than QJMR up to the 90%, this trend is consistent with QJMR's lower

overall meteor echo occurrence rate. The time intervals for both systems remain within 2.3 s for 99% of

Table 1. Cumulative distribution of meteor trail decay time and inter-arrival time for the SJMR and QJMR radar systems from 13th December 2024 to 27th January 2025 (All in the unit of second/s).

Percentile	SJMR (Decay/Interval)	QJMR (Decay/Interval)
50%	0.10/0.12	0.10/0.18
80%	0.19/0.31	0.17/0.45
90%	0.27/0.50	0.24/0.68
99%	0.70/1.92	0.69/2.24

the cases, suggesting that MBC systems can maintain relatively continuous communication. In summary, the short meteor trail inter-arrival times and brief decay durations impose stricter demands on the rapid response capability between the transmitter and receiver in a MBC system.

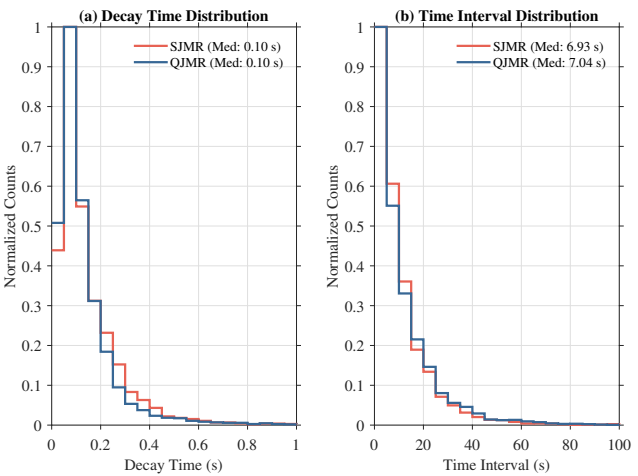


Figure 5. Combined statistical distributions of meteor trail decay time and inter-arrival time for the QJMR and SJMR systems, detected by the QJMR and SJMR from 13th December 2024 to 27th January 2025. The median values are provided.

To further investigate the spatiotemporal variations of these two parameters, we statistically analyze the meteor echoes observed by the SJMR and QJMR from 13th December 2024 to 27th January 2025 as shown in Figure 6. Both parameters exhibit distinct diurnal patterns that approximately follow Gaussian distributions from Figure 6(a) and (b). A notable increase in both decay time and inter-arrival time is observed during the local afternoon. Throughout this period, the decay times recorded by SJMR are consistently longer than those of QJMR. Furthermore, the inter-arrival time shows a clear inverse relationship with meteor

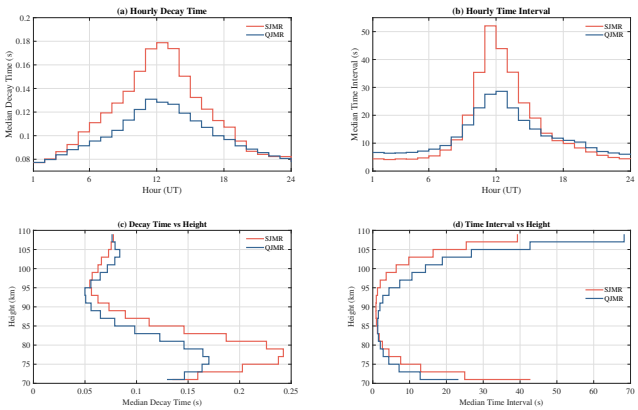


Figure 6. Spatiotemporal variations of meteor trail parameters observed by the SJMR and QJMR from 13th December 2024 to 27th January 2025. (a) Hourly decay time. (b) Hourly time interval. (c) Decay time versus specular height (d) Time interval versus specular height.

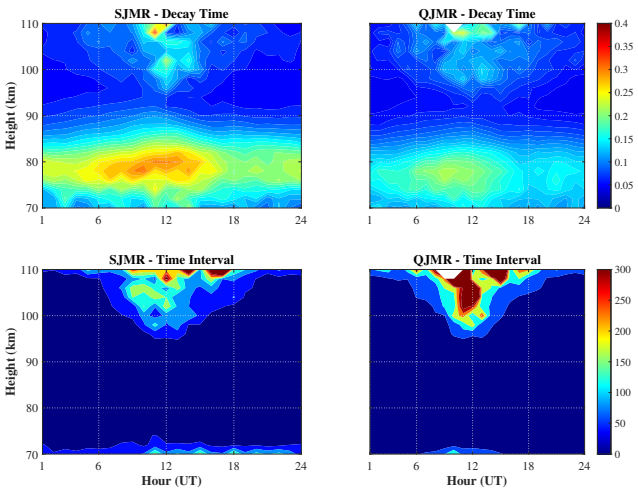


Figure 7. Meteor trail decay time and intervals time versus hour and specular height observed by the SJMR (left) and QJMR (right) from 13th December 2024 to 27th January 2025.

trail frequency, this is consistent with the findings in Figure 3(b), shorter during high-activity periods and longer when activity is low. Figure 6(c) reveals a non-monotonic relationship between decay time and specular height: it is relatively large at lower altitudes, decreases with increasing altitude, and then increases again. This pattern is closely linked to the altitudinal profile of the atmospheric ambipolar diffusion coefficient, which governs the meteor trail decay process. Furthermore, the time interval with specular height in Figure 6(d) is governed by the meteor trail echo occurrence rate. This is similar to Figure 3(d). Combining these spatiotemporal characteristics, Figure 7 reveals

the two-dimensional distributions of decay time and time interval in height-time space, which are governed by the combined effects of the background atmospheric ambipolar diffusion coefficient and the meteor trail echo occurrence rate.

The quantified meteor trail parameters, namely the distributions of trail decay time and inter arrival time as obtained from our radar observations, impose complementary and stringent constraints on media access control (MAC) protocol design. The statistical characteristics of these parameters directly inform both the fundamental feasibility and the practical optimization strategies for short range MBC systems. First, the extremely short trail decay time, with a median value of approximately 0.1 s, imposes a strict upper bound on the usable transmission window. This fundamental constraint necessitates that communication protocols exhibit minimal control overhead and achieve link establishment within millisecond timescales to utilize a meaningful portion of each trail's ephemeral lifetime. Second, the inter-arrival time distribution is characterized by a high density of very short intervals, with median values of 6.93 s and 7.04 s for the SJMR and QJMR systems, respectively. This common bursty temporal structure for communication opportunities necessitates protocols capable of rapidly resolving channel access contention to effectively exploit successive bursts of meteor trails. Furthermore, the nuanced differences in these statistical profiles between the two stations underscore the importance of considering site-specific channel statistics for protocol optimization and system deployment planning.

These distributions reveal specific limitations of conventional approaches. Basic ALOHA protocols, with their fixed and often long re-transmission delays, are particularly ill-suited as they would fail to capitalize on the prevalent short-interval opportunities, leading to poor channel utilization. Similarly, classical carrier sense multiple access (CSMA) protocols, which rely on detecting a persistent carrier, face fundamental challenges in an intermittently available channel like the meteor burst environment. This highlights the need for protocols specifically designed or significantly adapted for the unique temporal dynamics of the meteor channel. The pronounced diurnal variation evident in these parameters, as documented in Figure 6, mandates the adoption of adaptive, time-aware medium access strategies. During morning periods of elevated meteor ac-

tivity, which are characterized by minimal inter-arrival times, protocol parameters such as contention window size should be configured to facilitate aggressive channel access, thereby maximizing aggregate throughput. Conversely, during afternoon periods where observations indicate a tendency toward extended trail decay times, the strategic focus should shift to optimizing the data payload of individual bursts. This can be achieved through mechanisms such as adaptive frame length control or the allocation of contention-free transmission slots, ensuring efficient utilization of the prolonged yet less frequent communication windows.

Critically, the combination of brief trail lifetimes and highly frequent occurrences creates a channel with numerous, short-duration communication windows. The fact that the system's parameter acquisition time (0.66–2 ms) is orders of magnitude shorter than the median trail duration ensures that a significant portion of each window is available for data exchange. Consequently, the dense succession of these usable bursts enables MBC systems to sustain macroscopically near-continuous communication, provided the MAC protocol is sufficiently agile to match the channel's temporal dynamics.

V. SNR CHARACTERISTICS OF METEOR TRAILS

Signal-to-noise ratio (SNR) is a critical metric for assessing the quality and availability of meteor burst communication (MBC) links. In this section, we analyze the SNR distributions observed by the SJMR (39.0 MHz) and QJMR (53.1 MHz) systems and discuss their implications for short-range MBC system design. Figure 8 (a) illustrates the diurnal variation of hourly median SNR for both radar systems. Both systems exhibit a clear diurnal cycle in SNR, with similarly low levels observed between 08:00 and 14:00 UT. However, their temporal variation patterns are markedly different. The SJMR system consistently records a higher overall SNR compared to QJMR, but it also shows significantly sharper and more abrupt fluctuations. In contrast, the QJMR system, while operating at a lower absolute SNR, demonstrates a notably more stable temporal profile across the diurnal cycle. This suggests that although the lower-frequency system benefits from enhanced signal quality, it is also more susceptible to pronounced temporal variability. Figure 8 (b) presents

the distribution of median SNR as a function of meteor trail specular height. Both systems show a peak SNR near 95 km, which is slightly different from the altitude of maximum meteor echo occurrence shown in Figure 3.

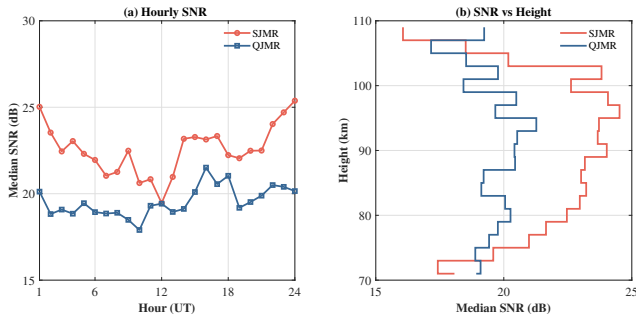


Figure 8. Spatiotemporal variations of meteor trail SNR observed by the SJMR and QJMR from 13th December 2024 to 27th January 2025. (a) Hourly SNR. (b) SNR versus specular height.

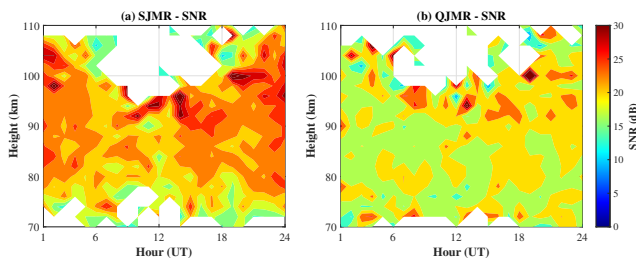


Figure 9. Meteor trail SNR versus hour and specular height observed by the SJMR (left) and QJMR (right) from 13th December 2024 to 27th January 2025.

Given that the vast majority of meteor trails are underdense trails and extremely short decay time, a high SNR is essential to ensure successful signal detection and data decoding within the brief communication window. The consistently higher SNR of the SJMR system suggests that lower frequency is more favorable for maximizing link availability. The pronounced diurnal and specular height variations in SNR, especially as showed in the Figure 9, suggest that MBC systems could significantly benefit from intelligent, spatiotemporally aware transmission scheduling. Communication sessions prioritized during the high-SNR window and focused on the 95 km altitude layer would experience maximized link robustness. Furthermore, the observed SNR distributions provide a basis for optimizing antenna gain patterns and receiver noise figures. Systems operating at higher frequencies, for example,

may require higher-gain antennas or lower-noise amplifiers to compensate for their inherent SNR disadvantage and more restricted high-quality spatiotemporal domain. In summary, the SNR characteristics provide valuable empirical constraints for MBC system optimization. By integrating these SNR statistics with the previously discussed distributions of meteor trail occurrence, decay time, and inter-arrival time, MBC designers can develop more accurate channel models and thereby enhance the overall robustness and availability of short-range meteor burst communication systems.

VI. CONCLUSION

This paper presents a comprehensive statistical analysis of meteor trail characteristics using concurrent observations from two back-scatter meteor radars (SJMR and QJMR) in Yunnan, China, from 13th December 2024 to 27th January 2025. Our investigation reveals fundamental spatiotemporal patterns and physical parameters of underdense meteor trails that have direct implications for the MBC system design.

The analysis demonstrates significant frequency-dependent characteristics in meteor trail detection, where the lower-frequency SJMR detected substantially more meteor trails (averaging 7842 daily) compared to the higher-frequency QJMR (6153 daily), indicating generally better communication opportunities at lower frequencies. However, diurnal variations reveal a notable reversal during local afternoon hours when the higher-frequency system exhibits superior communication potential, suggesting the need for frequency-adaptive scheduling in MBC operations. Spatial analysis shows that meteor trails detected by lower-frequency radar experience longer propagation paths and consequently greater path loss, providing crucial guidance for frequency selection in MBC modeling, while the near-uniform azimuthal distribution supports omnidirectional antenna configurations, though site-specific hardware constraints must be considered.

Analysis of meteor trail decay time characteristics further demonstrate MBC system feasibility. Although lower-frequency radar demonstrates slightly longer decay times, both systems exhibit remarkably short durations (generally within 0.10 s). Critically, the rapid parameter acquisition time (0.66–2 ms) is orders of magnitude shorter than the typical trail duration, ensuring sufficient time for communication establishment

and data transmission. Furthermore, the observed anti-correlation between meteor echo occurrence rate and inter-arrival time reveals an important operational trade-off: during local afternoon periods of reduced meteor activity, extended waiting times between communication opportunities are compensated by longer trail durations that permit larger data transmissions per burst.

Future studies should extend the observational analysis to seasonal time scales to better characterize long-term channel variability. Additionally, the protocol design implications suggested by our statistical findings require validation through simulation, using the empirical distributions presented here as benchmarks for evaluating adaptive MAC strategies. In summary, this research provides authentic physical parameters essential for realistic MBC channel modeling, offering scientifically grounded references for system optimization in frequency selection, temporal scheduling, and antenna configuration.

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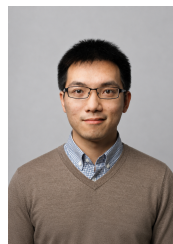
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